Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 440828388

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.135074 d
  Epoch (t0)    1492.2944 BTJD
  Duration      0.0840 d
  Depth         4.038 %  (40,380 ppm)
  SNR           165.7

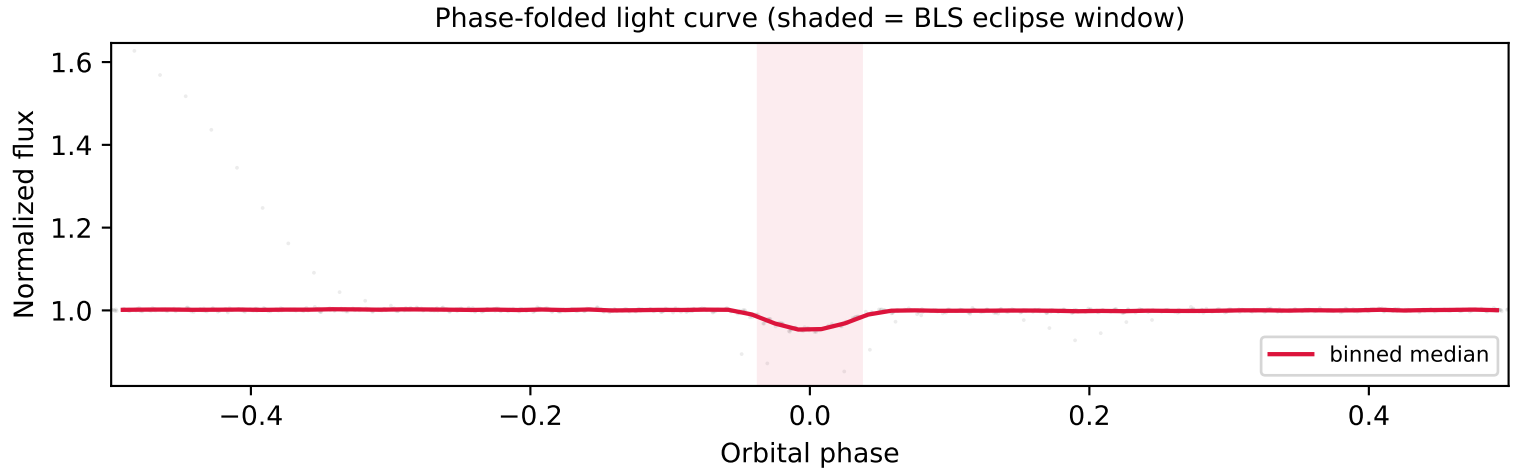
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      20  (gate >= 3)
  Robust depth ratio     0.94  (gate >= 0.50)
  Robust depth sigma     100.2  (gate >= 7.0)
  Odd/even z-score       0.51  (consistent)
  Secondary eclipse      not detected
```

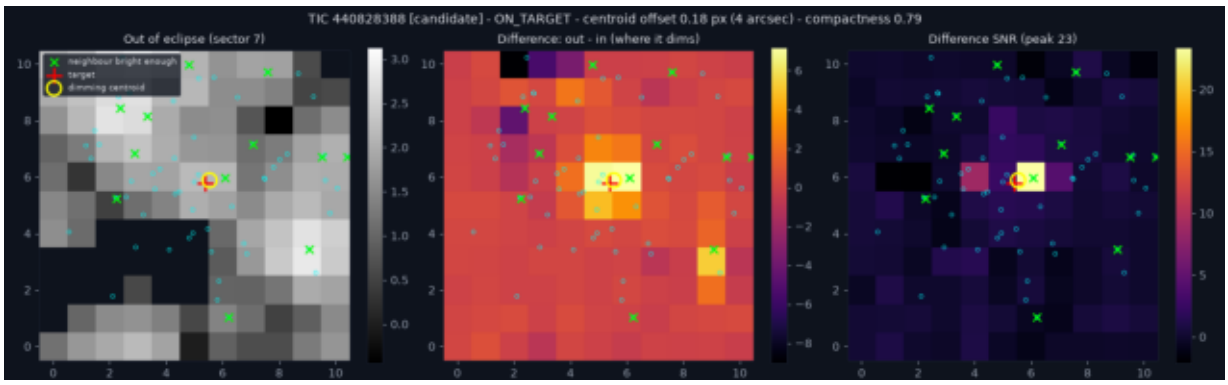
```
CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 87.9 sigma    t0 refined by -0.0150 phase
  Difference-image peak SNR   23.43   (gate >= 5.0)      Compactness   0.79   (gate >= 0.50)
  Dimming centroid offset     0.18 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 440828391
    0.54 px away, dTmag +3.10 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 389751698

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       10.644713 d  (ALIASED, see below)
    -> corrected period: 21.289427 d
  Epoch (t0)   1354.9147 BTJD
  Duration     0.1200 d
  Depth        19.536 %  (195,362 ppm)
  SNR          142.3

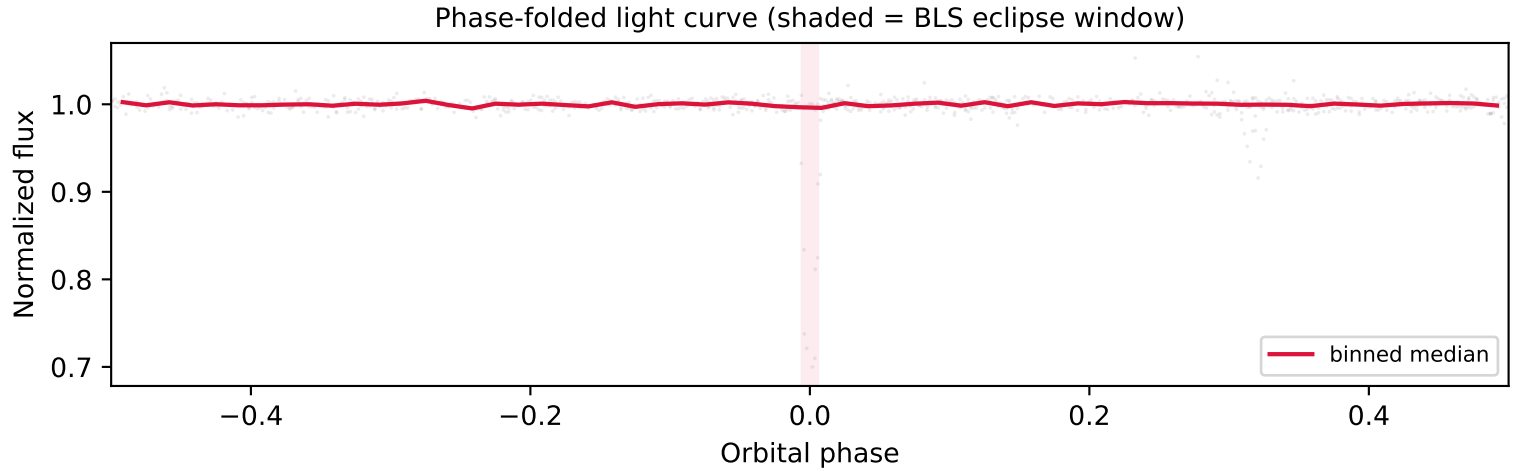
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     3  (gate >= 3)
  Robust depth ratio    0.97  (gate >= 0.50)
  Robust depth sigma    106.7  (gate >= 7.0)
  Odd/even z-score      63.20  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

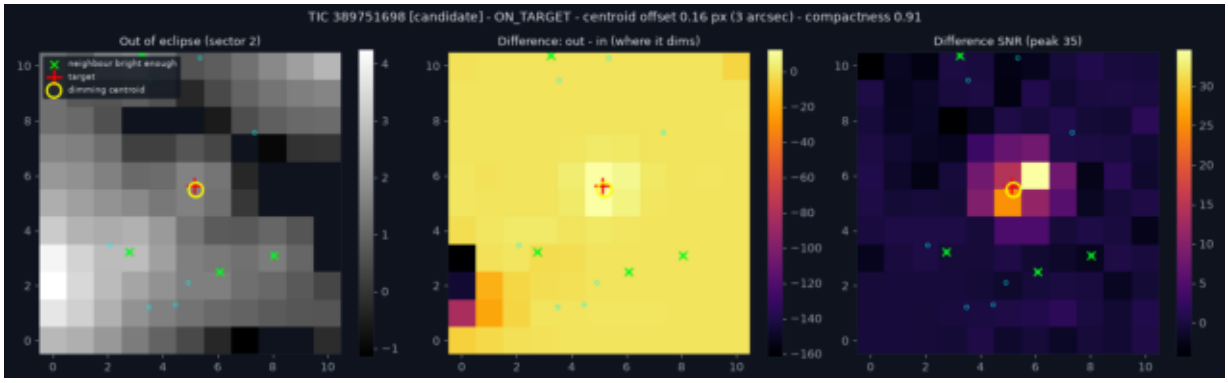
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 2    light-curve dip in this sector: 139.2 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   34.64   (gate >= 5.0)      Compactness   0.91   (gate >= 0.50)
  Dimming centroid offset     0.16 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 389728981
    3.11 px away, dTmag +0.55 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 63.20 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 21.289427 d, in any write-up.

TIC 237657109

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       2.545124 d
  Epoch (t0)   1469.5678 BTJD
  Duration     0.1700 d
  Depth        1.553 %  (15,529 ppm)
  SNR          56.3

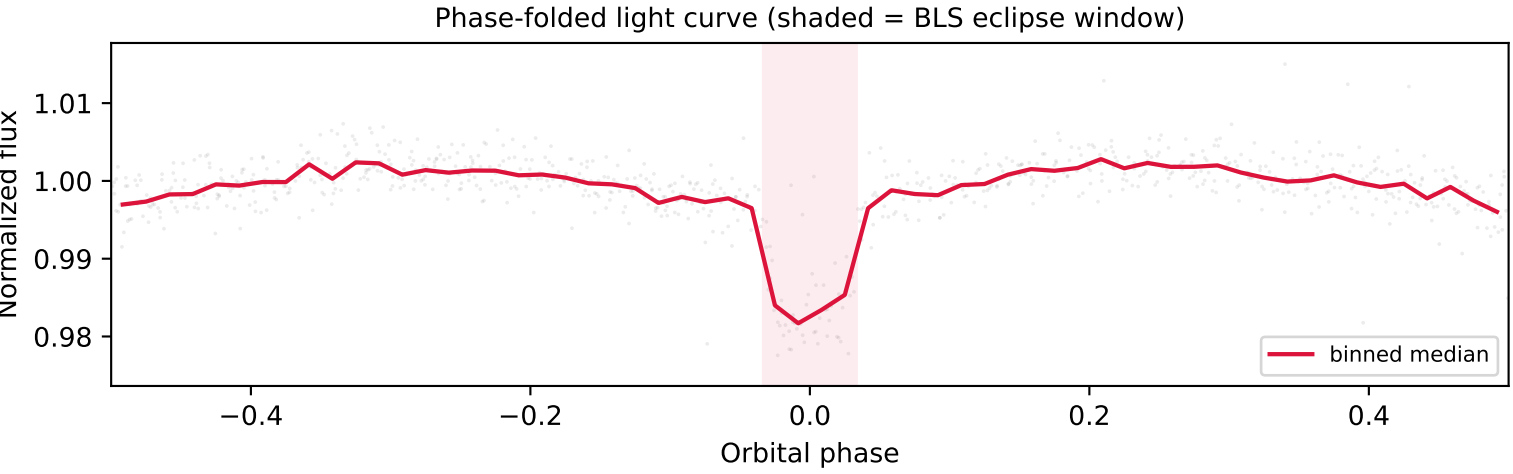
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     8  (gate >= 3)
  Robust depth ratio    1.08  (gate >= 0.50)
  Robust depth sigma    36.4  (gate >= 7.0)
  Odd/even z-score      1.72  (consistent)
  Secondary eclipse     not detected
```

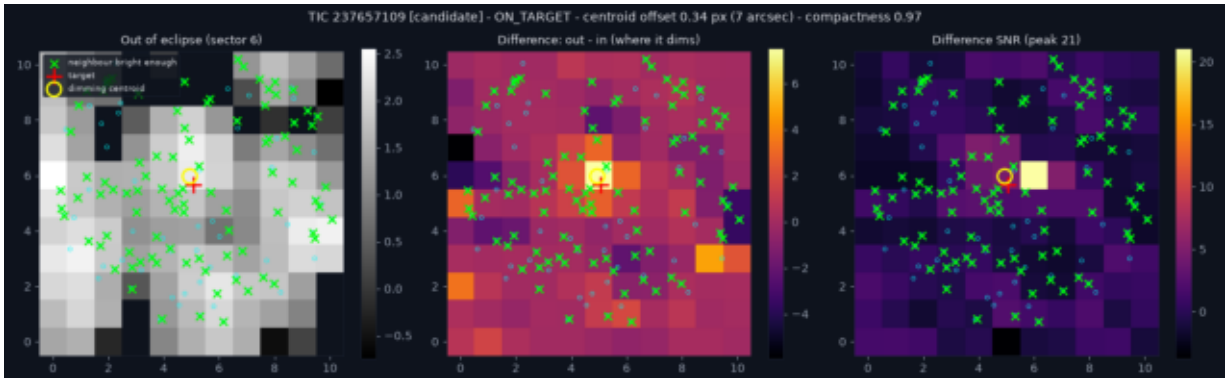
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 6    light-curve dip in this sector: 49.6 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   21.09   (gate >= 5.0)      Compactness   0.97   (gate >= 0.50)
  Dimming centroid offset     0.34 px (7 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 237657122
    0.48 px away, dTmag -1.08 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 403329469

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       30.939496 d  (ALIASED, see below)
    -> corrected period: 61.878992 d
  Epoch (t0)   2838.1348 BTJD
  Duration     0.5000 d
  Depth        0.051 %  (511 ppm)
  SNR          19.2

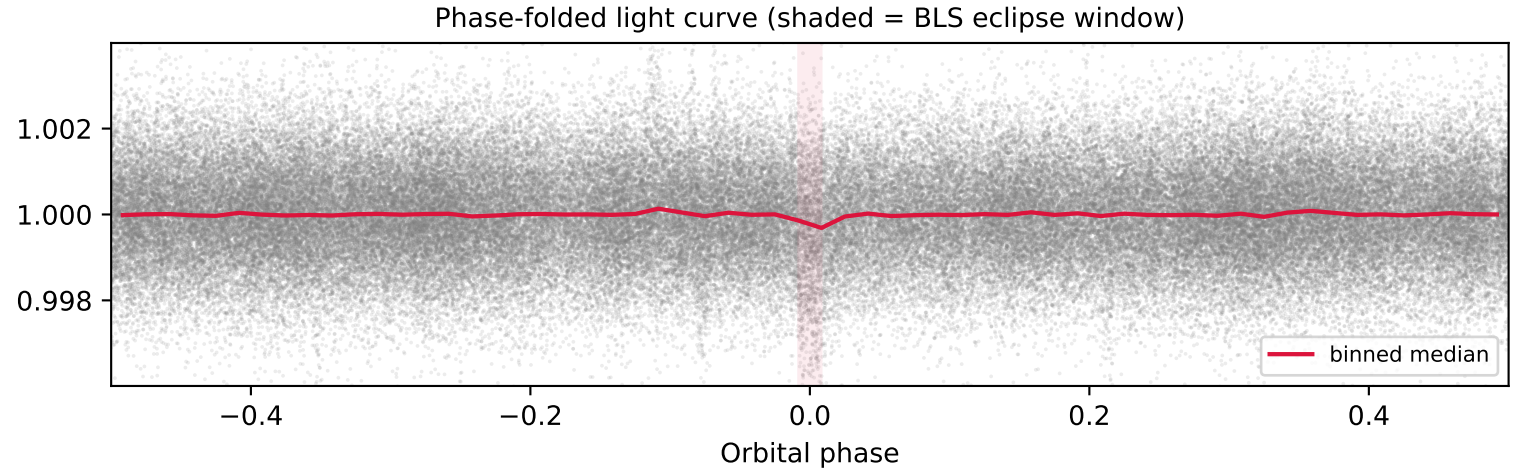
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     8  (gate >= 3)
  Robust depth ratio    0.69  (gate >= 0.50)
  Robust depth sigma    10.4  (gate >= 7.0)
  Odd/even z-score      20.32  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

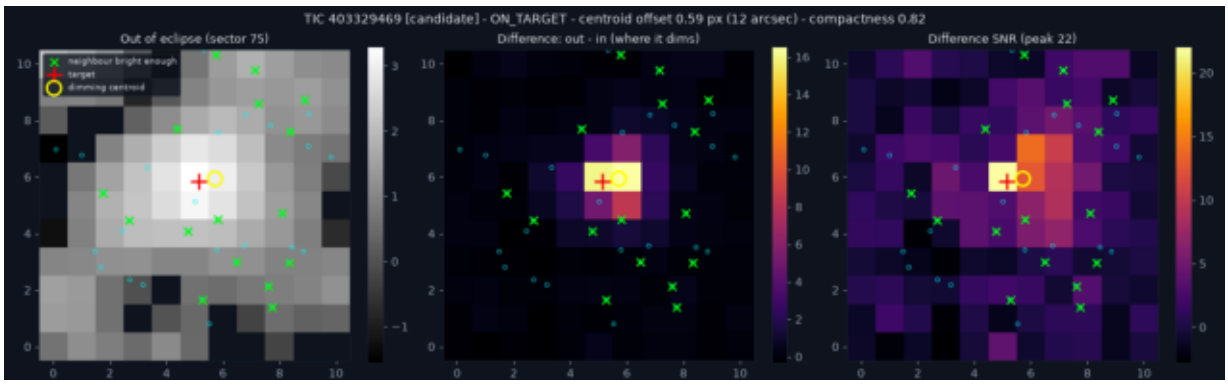
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 75    light-curve dip in this sector: 9.9 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   22.05   (gate >= 5.0)      Compactness   0.82   (gate >= 0.50)
  Dimming centroid offset     0.59 px (12 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 403333340
    1.45 px away, dTmag +7.78 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 20.32 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 61.878992 d, in any write-up.